Model Question Paper-1 with effect from 2019-20 (CBCS Scheme)

USN | | | | | | | | | | |

Fifth Semester B.E. Degree Examination

Data Base Management System

TIME: 03 Hours **Max. Marks: 100**

Note: 01. Answer any **FIVE** full questions, choosing at least **ONE** question from each **MODULE**.

		Module – 1	
Q.1	**(a)**	Compare DBMS and early file systems bringing out the major advantages of the database approach.	6M
	(b)	With a neat block diagram, explain the architecture of a typical DBMS.	10M
	(c)	Explain different types of user friendly interfaces and types of user who typically use each.	4M
		OR	
Q2	**(a)**	Define the following with an example: (i) Weak entity type (ii) participation constraint (iii)cardinality ratio (iv) recursive relationship (v) specialization	10M
	(b)	Develop an ER diagram for keeping track of information about a company database taking into account atleast five entities.	10M
		Module – 2	
Q.3	**(a)**	Define the following terms i) Key ii) Super key iii) Candidate key iv) Primary key v) Foreign key	5M
	(b)	Consider the following COMPANY database EMP(Name,SSN,Salary,SuperSSN,Gender,Dno) DEPT(DNum,Dname,MgrSSN,Dno) DEPT_LOC(Dnum,Dlocation) DEPENDENT(ESSN,Dep_name,Sex) WORKS_ON(ESSN,Pno,Hours) PROJECT(Pname,Pnumber,Plocation,Dnum) Write the **relational algebra** queries for the following (i)Retrieve the name, address, salary of employees who work for the Research department. (ii) find the names of employees who work on all projects controlled by department number4. iii) Retrieve the SSN of all employees who either in department no :4 or directly supervise an employee who work in department number :4 (iv) Retrieve the names of employees who have no dependents (v) Retrieve each department number, the number of employees in the department and their average salary.	10M
	(c)	Summarize the steps involved in converting the ER constructs to corresponding relational tables.	5M
		OR	
	(a)	Explain with example basic constraints that can be specified when a database table is created.	4M

Q.4	(b)	Write SQL syntax for the following with example: (i) SELECT (ii) ALTER (iii) UPDATE	6
	(c)	Consider the following relation schema Works(Pname,Cname,salary) Lives(Pname,Street,City) located_in (Cname, city) Manager(Pname,Mgrname) Write the SQL queries for the following i) Find the names of all persons who live in the city Bangalore. ii) Retrieve the names of all person of "Infosys" whose salary is between Rs .50000 iii)Find the names of all persons who lives and work in the same city iv)List the names of the people who work for "Tech M" along with the cities they live in. v)Find the average salary of "Infosys" persons	10M
Module – 3			
Q.5	(a)	Explain the following constructs used in SQL with example: (i)Nested queries ii)Aggregate functions iii) Triggers iv)Views and their updability v) schema change statements	5M
	(b)	What is Dynamic SQL? How it is different from Embedded SQL	10M
	(c)	Consider the following COMPANY database(10 marks) EMP(Name,SSN,Salary,SuperSSN,Dno) DEPT(DNum,Dname,MgrSSN,Dno) DEPT_LOC(Dnum,Dlocation) DEPENDENT(ESSN,Dep_name,Sex) WORKS_ON(ESSN,Pno,Hours) PROJECT(Pname,Pnumber,Plocation,Dnum) Write the **SQL** queries for the following i) Retrieve the name of the employee who works with same department as ravi ii)Retrieve the number of dependents for an employee "Ravi" iii)Retrieve the name of the managers working in location "DELHI"who has no female dependents iv)List female employees from Dno=20 earning more than 50000 v)List "CSE" department details	5M
OR			
Q.6	(a)	What is SQLJ? How it is different from JDBC.	4M
	(b)	Draw and explain 3-tier Architecture and technology relevant to each tier. Write the advantages of 3- tier architecture.	6M
	(c)	Write a short note on a)html forms b)Java scripts c)CGI d)Application servers e)Servlets	10M
Module – 4			
Q.7	(a)	Explain insertion, deletion and modification anomalies. Why are they considered bad? Illustrate with example.	4M
	(b)	Explain 1NF,2NF,3NF with example.	6M
	(c)	Consider the relation schema R(A,B,C,D,E,F) and the functional dependencies A->B,C->DF,AC->E, D->F.What is the primary key of this relation R?What is its highest normal form?Preserving the dependency,decompose R into third normal form.	10M
OR			

Q.8	**(a)**	Define non-additive join property of a decomposition and write an algorithm for testing of non additive join property.	5M
	(b)	Write the algorithm to find the minimal cover for a sets of FD's Consider R={A,B,C,D,E,F} .FD's {A->C,AC->D,E->AD,E->H} Find the irreducible cover for this set of FD's(minimal cover)	10M
	(c)	Given below are two sets of FD's for a relation R(A,B,C,D,E).Are they equivalent? F={A->C,AC->D,E->AD,E->H} and G={A->CD,E->AH}	5M
Module – 5			
Q.9	**(a)**	Briefly discuss on the two phase locking protocol used in concurrency control. How does it gurantees serializability.	10M
	(b)	Check whether given schedule is serializable or not using precedence graph. Explain with algorithm. S1:R1(X) R2(Z) R1(Z) R3(X) R3(Y) W1(X) W3(Y) R2(Y) W2(Z) W2(Y)	6M
	(c)	Explain Basic time stamping algorithm.	4M
OR			
Q.10	**(a)**	Explain multi version concurrency control protocols.	10M
	(b)	Write short notes on the following I. transaction rollback and cascading rollback. II. transaction support in SQL III. shadow paging IV. NO-UNDO/REDO Recovery Based on Deferred Update V. Recovery Techniques Based on Immediate Update	10M